

Rommel Jalasutram

SENIOR SOFTWARE DEVELOPMENT ENGINEER · DISTRIBUTED SYSTEMS · MACHINE LEARNING · DEEP LEARNING

Lasalle, ON, Canada

✉ +1 519-984-3555 ✉ r.jalasutram@proton.me ✉ rjalasutram ✉ Rommel Jalasutram

Profile

Software engineer with a Ph.D. in Computer Science and 10+ years of experience designing and building large-scale distributed and machine learning systems across ad tech and cloud platforms. Expertise in data engineering, machine learning for ad auctions and forecasting, and GPU-accelerated model training and inference. Delivered systems processing billions of clicks and impressions daily, forecasting page views and impressions, and optimizing campaign budget allocation. **Canadian Citizen**

Work Experience

Criteo Corp Canada

Remote

SENIOR SOFTWARE DEVELOPMENT ENGINEER

Dec 2020 – Apr 2026

Worked across data engineering and applied machine learning teams, processing billions of clicks, impressions, page views, and other advertising events.

Ad Auction Design

- Architected a budget allocation mechanism that distributes daily campaign budgets across line items based on historical spending patterns, improving budget utilization.
- Designed a real-time budget pacing mechanism to improve both budget utilization and time-to-live for line items; contributed to a PID controller-based approach for real-time feedback.
- Led the design and development of components to support billions of auctions conducted daily for placement impression based ads while meeting to sub-millisecond latency requirements.
- Led the design and development of Holistic Page Optimization, enabling organic and sponsored products to be ranked and displayed together while optimizing for retailer-defined objectives.
- *Tech stack:* Python, C#, Spark, Scala, Flink.

GPU Optimization

- Increased deep learning training throughput by 10% by optimizing data movement between HDFS and GPUs.
- Increased GPU utilization from 85% to 98% through experiments with batch sizing, quantization, and data compression.
- *Tech stack:* Python, Ray, PyTorch.

Data Engineering & Forecasting

- Built hourly and real-time data pipelines processing billions of clicks, impressions, page views, and other advertising events, enabling data transformations for reporting, machine learning, and analytics.
- Developed an automated system to continuously replicate database contents to HDFS while detecting and tracking schema changes across multiple databases.
- Led the design and development of a Forecasting API service for estimating placement impressions.
- *Tech stack:* Python, Scala, XGBoost.

Amazon.com Services LLC Inc

Detroit, MI, USA

SOFTWARE DEVELOPMENT ENGINEER II

Apr 2017 – Dec 2020

Worked in the Marketplace Web Services Team

- Designed, developed, and scaled distributed backend services handling more than one billion requests per day.
- Designed and developed a secure, reliable, and operationally efficient platform for serving marketplace resources to sellers, developers, and authorized customers.
- Designed and implemented components for authentication, authorization, throttling, metering, error reporting, monitoring, and request transformation.
- Built developer tooling using Git libraries and AWS to automate documentation generation and streamline onboarding of backend teams, reducing a manual process from three days to a few minutes.
- *Tech stack:* Java, AWS.

NetApp Inc

Research Triangle Park, NC, USA

MEMBER OF TECHNICAL STAFF III

Oct 2014 - Mar 2017

- Developed and maintained C and C++ software across user and kernel space as a member of the ONTAP OS kernel team, improving high-availability clustering and disk performance.
- Contributed to the design and implementation of continuous metadata block mirroring to reduce takeover times between partner nodes, achieving a 20% reduction in takeover time across multiple WAFL filesystem components.
- Maintained and enhanced disk layout features, including Segment Cleaning and Free Space Reallocation, as well as Flash Cache functionality.
- *Tech stack:* C, C++.

Infosys Technologies Ltd

Mysore, India

SOFTWARE ENGINEER

Aug 2005 – Aug 2007

- Worked in projects related to insurance and banking domains. This involved developing DLLs and serviced components for various modules using VB DoTNET and SQL server.
- Lead a team of five people to test the check processing infrastructure for a financial provider in Canada. Completed the project in time and helped the customer find fatal flaws.
- *Tech stack:* C++, VB.NET.

Selected Publications

Google Scholar

- Brian C. Dean, and Rommel Jalasutram. **Factor Revealing LPs and Stable Matching with Ties and Incomplete Lists.** *MATCH-UP*, April 2015
- Brian C. Dean, Rommel Jalasutram, and Chad Waters. **Lightweight Approximate Selection.** *European Symposium on Algorithms(ESA)*, September 2014.
- Pavan Yalamanchili, Sumod K Mohan, Rommel Jalasutram, and Tarek Taha. **Acceleration of Hierarchical Bayesian Network Based Cortical Models on Multicore Architectures.** *Elsevier Parallel Computing*, 2010.
- Tarek Taha, Pavan Yalamanchili, Mohammed Bhuiyan, Rommel Jalasutram, Chong Chen, and Richard Linderman. **Neuromorphic Algorithms on Clusters of PlayStation 3s.** *International Joint Conference on Neural Networks(IJCNN)*, July 2010.

Education

Clemson University

Clemson, SC, USA

- PH.D., COMPUTER SCIENCE
- DISSERTATION: AN APPROXIMATION ALGORITHM FOR THE STABLE MARRIAGE PROBLEM WITH TIES AND INCOMPLETE LISTS
- ADVISOR: DR. BRIAN C. DEAN

2014

Clemson University

Clemson, SC, USA

- M.S., COMPUTER ENGINEERING
- DISSERTATION: ACCELERATION OF SPIKING NEURAL NETWORKS ON MULTICORE ARCHITECTURES
- ADVISOR: DR. TAREK M. TAHA

2009

National Institute of Technology

Surat, India

B.TECH., COMPUTER ENGINEERING

2005

Skills

Languages

Python, C, C++, Scala

ML Libraries

PyTorch, Ray, TensorFlow, GPU Programming

Big Data

Spark, Flink

Deployment

TensorRT, ONNX, Quantization, Edge Inference

Domains

Ad Tech, Cloud Computing